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Imaging lens assembly and electronic device

Abstract

An imaging lens assembly includes a first lens element, a second lens element and a lens barrel, and an optical axis passes through the imaging lens assembly. One of the space adjusting structures is formed via a first peripheral portion of the first lens element and a plate portion of the lens barrel, the other one of the space adjusting structures is formed via the first peripheral portion of the first lens element and a second peripheral portion of the second lens element. Each of the space adjusting structures includes a frustum surface, a spatial frustum surface, a corresponding structure and a spatial layer. Each of the frustum surfaces and each of the spatial frustum surfaces are disposed on an object-side surface of the first peripheral portion and an object-side surface of the second peripheral portion, respectively.

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application Ser. No. 63/302,588, filed Jan. 25, 2022 and Taiwan Application Serial Number 111125766, filed Jul. 8, 2022, which are herein incorporated by reference.

BACKGROUND
Technical Field
(1) The present disclosure relates to an imaging lens assembly. More particularly, the present disclosure relates to an imaging lens assembly applicable to portable electronic devices.
Description of Related Art
(2) In recent years, portable electronic devices have developed rapidly. For example, intelligent electronic devices and tablets have been filled in the lives of modern people, and imaging lens assemblies mounted on portable electronic devices have also prospered. However, as technology advances, the quality requirements of the imaging lens assembly are becoming higher and higher. Therefore, an imaging lens assembly, which can maintain the assembling stability under the different environmental conditions, needs to be developed.
SUMMARY
(3) According to one aspect of the present disclosure, an imaging lens assembly includes a first lens element, a second lens element, a lens barrel and two space adjusting structures, and an optical axis passes through the imaging lens assembly. The first lens element includes a first optical effective portion and a first peripheral portion. The optical axis passes through the first optical effective portion, and the first peripheral portion is disposed around the first optical effective portion. The second lens element is disposed on an image side of the first lens element, and includes a second optical effective portion and a second peripheral portion. The optical axis passes through the second optical effective portion, the second peripheral portion is disposed around the second optical effective portion, and an object-side surface of the second peripheral portion is directly contacted with an image-side surface of the first peripheral portion. The lens barrel includes a cylindrical portion and a plate portion. The cylindrical portion surrounds the optical axis with the optical axis as an axis, the plate portion is connected to the cylindrical portion, extends towards a direction close to the optical axis to form a light through hole, an accommodating space is formed via the cylindrical portion and the plate portion, the first lens element and the second lens element are disposed in the accommodating space, and an image-side surface of the plate portion is directly contacted with an object-side surface of the first peripheral portion. One of the space adjusting structures is formed via the first peripheral portion of the first lens element and the plate portion of the lens barrel, and the other one of the space adjusting structures is formed via the first peripheral portion of the first lens element and the second peripheral portion of the second lens element. The one of the space adjusting structures includes a frustum surface, a spatial frustum surface, a corresponding structure and a spatial layer. The frustum surface is disposed on the object-side surface of the first peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is closer to the optical axis than an image-side end of the frustum surface to the optical axis. The spatial frustum surface is disposed on the object-side surface of the first peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface is farther from the optical axis than an image-side end of the spatial frustum surface from the optical axis. The corresponding structure is disposed on the image-side surface of

the plate portion and correspondingly disposed on the frustum surface and the spatial frustum surface. The spatial layer is formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure are disposed at intervals. The other one of the space adjusting structures includes a frustum surface, a spatial frustum surface, a corresponding structure and a spatial layer. The frustum surface is disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is farther from the optical axis than an image-side end of the frustum surface from the optical axis. The spatial frustum surface is disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface is closer to the optical axis than an image-side end of the spatial frustum surface to the optical axis. The corresponding structure is disposed on the image-side surface of the first peripheral portion and correspondingly disposed on the frustum surface and the spatial frustum surface. The spatial layer is formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure are disposed at intervals. When the imaging lens assembly is in a first environment, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the space adjusting structures is $G\alpha$, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the space adjusting structures is $G\beta$; when the imaging lens assembly is in a second environment, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the space adjusting structures is $G\alpha'$, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the space adjusting structures is $G\beta'$, and the following conditions are satisfied: $0\ \mu m \leq G\alpha' < G\alpha \leq 37\ \mu m$; and $0\ \mu m \leq G\beta' < G\beta \leq 38\ \mu m$. The first environment and the second environment are satisfied at least one of a temperature-dependent relation and a humidity-dependent relation: a temperature of the first environment being Ta, a temperature of the second environment being Tb, and the temperature-dependent relation satisfied: $6K \leq |Ta-Tb| \leq 148K$; and a relative humidity of the first environment being RHa, a relative humidity of the second environment being RHb, and the humidity-dependent relation satisfied: $7\% \leq |RHa-RHb| \leq 89\%$.

(4) According to one aspect of the present disclosure, an imaging lens assembly includes a first lens element, a lens barrel and a space adjusting structure, and an optical axis passes through the imaging lens assembly. The first lens element includes a first optical effective portion and a first peripheral portion. The optical axis passes through the first optical effective portion, and the first peripheral portion is disposed around the first optical effective portion. The lens barrel includes a cylindrical portion and a plate portion. The cylindrical portion surrounds the optical axis with the optical axis as an axis, the plate portion is connected to the cylindrical portion, extends towards a direction close to the optical axis to form a light through hole, an accommodating space is formed via the cylindrical portion and the plate portion, the first lens element is disposed in the accommodating space, and an image-side surface of the plate portion is directly contacted with an object-side surface of the first peripheral portion. The space adjusting structure is formed via the first peripheral portion of the first lens element and the plate portion of the lens barrel, and the space adjusting structure includes a frustum surface, a spatial frustum surface, a corresponding structure and a spatial layer. The frustum surface is disposed on the object-side surface of the first peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is closer to the optical axis than an image-side end of the frustum surface to the optical axis. The spatial frustum surface is disposed on the object-side surface of the first peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface is farther from the optical axis than an image-side end of the spatial frustum surface from the optical axis. The corresponding structure is disposed on the image-side surface of the plate portion and correspondingly disposed on the frustum surface and the spatial frustum surface. The spatial layer

is formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure are disposed at intervals. When the imaging lens assembly is in a first environment, a minimum spacing distance between the spatial frustum surface and the corresponding structure is G; when the imaging lens assembly is in a second environment, the minimum spacing distance between the spatial frustum surface and the corresponding structure is G′, and the following condition is satisfied: $0\ \mu m \leq G' < G \leq 37\ \mu m$. The first environment and the second environment are satisfied at least one of a temperature-dependent relation and a humidity-dependent relation: a temperature of the first environment being Ta, a temperature of the second environment being Tb, and the temperature-dependent relation satisfied: $6K \leq |Ta-Tb| \leq 148K$; and a relative humidity of the first environment being RHa, a relative humidity of the second environment being RHb, and the humidity-dependent relation satisfied: $7\% \leq |RHa-RHb \leq| 89\%$.

(5) According to one aspect of the present disclosure, an imaging lens assembly includes a first lens element, a second lens element, a third lens element and two space adjusting structures, and an optical axis passes through the imaging lens assembly. The first lens element includes a first optical effective portion and a first peripheral portion. The optical axis passes through the first optical effective portion, and the first peripheral portion is disposed around the first optical effective portion. The second lens element is disposed on an image side of the first lens element, and includes a second optical effective portion and a second peripheral portion. The optical axis passes through the second optical effective portion, the second peripheral portion is disposed around the second optical effective portion, and an object-side surface of the second peripheral portion is directly contacted with an image-side surface of the first peripheral portion. The third lens element is disposed on an image side of the second lens element, and includes a third optical effective portion and a third peripheral portion. The optical axis passes through the third optical effective portion, the third peripheral portion is disposed around the third optical effective portion, and an object-side surface of the third peripheral portion is directly contacted with an image-side surface of the second peripheral portion. The one of the space adjusting structures is formed via the first peripheral portion of the first lens element and the second peripheral portion of the second lens element, and the other one of the space adjusting structures is formed via the second peripheral portion of the second lens element and the third peripheral portion of the third lens element. The one of the space adjusting structures includes a frustum surface, a spatial frustum surface, a corresponding structure and a spatial layer. The frustum surface is disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is farther from the optical axis than an image-side end of the frustum surface from the optical axis. The spatial frustum surface is disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface is closer to the optical axis than an image-side end of the spatial frustum surface to the optical axis. The corresponding structure is disposed on the image-side surface of the first peripheral portion and correspondingly disposed on the frustum surface and the spatial frustum surface. The spatial layer is formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure are disposed at intervals. The other one of the space adjusting structures includes a frustum surface, a spatial frustum surface, a corresponding structure and a spatial layer. The frustum surface is disposed on the object-side surface of the third peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is closer to the optical axis than an image-side end of the frustum surface to the optical axis. The spatial frustum surface is disposed on the object-side surface of the third peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface is farther from the optical axis than an image-side end of the spatial frustum surface from the optical axis. The corresponding structure is disposed on the image-side surface of the second peripheral portion and correspondingly disposed on the frustum surface and the spatial frustum surface. The spatial layer is formed between the spatial frustum surface and the

corresponding structure, so that the spatial frustum surface and the corresponding structure are disposed at intervals. When the imaging lens assembly is in a first environment, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the space adjusting structures is $G\gamma$, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the space adjusting structures is $G\delta$; when the imaging lens assembly is in a second environment, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the space adjusting structures is $G\gamma'$, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the space adjusting structures is $G\delta'$; an abbe number of the second lens element is Vd, and the following conditions are satisfied: $3\ \mu m \leq G\gamma' < G\gamma \leq 38\ \mu m$; $3\ \mu m \leq G\leq' < G\delta \leq 39\ \mu m$; and $8 \leq Vd \leq 29$. The first environment and the second environment are satisfied at least one of a temperature-dependent relation and a humidity-dependent relation: a temperature of the first environment being Ta, a temperature of the second environment being Tb, and the temperature-dependent relation satisfied: $6K \leq |Ta-Tb| \leq 148K$; and a relative humidity of the first environment being RHa, a relative humidity of the second environment being RHb, and the humidity-dependent relation satisfied: $7\% \leq |RHa-RHb| \leq 89\%$.

(6) According to one aspect of the present disclosure, an imaging lens assembly includes a first lens element, a second lens element, a third lens element and two space adjusting structures, and an optical axis passes through the imaging lens assembly. The first lens element includes a first optical effective portion and a first peripheral portion. The optical axis passes through the first optical effective portion, and the first peripheral portion is disposed around the first optical effective portion. The second lens element is disposed on an image side of the first lens element, and includes a second optical effective portion and a second peripheral portion. The optical axis passes through the second optical effective portion, the second peripheral portion is disposed around the second optical effective portion, and an object-side surface of the second peripheral portion is directly contacted with an image-side surface of the first peripheral portion. The third lens element is disposed on an image side of the second lens element, and includes a third optical effective portion and a third peripheral portion. The optical axis passes through the third optical effective portion, the third peripheral portion is disposed around the third optical effective portion, and an object-side surface of the third peripheral portion is directly contacted with an image-side surface of the second peripheral portion. One of the space adjusting structures is formed via the first peripheral portion of the first lens element and the second peripheral portion of the second lens element, and the other one of the space adjusting structures is formed via the second peripheral portion of the second lens element and the third peripheral portion of the third lens element. The one of the space adjusting structures includes a frustum surface, a spatial frustum surface, a corresponding structure and a spatial layer. The frustum surface is disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is farther from the optical axis than an image-side end of the frustum surface from the optical axis. The spatial frustum surface is disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface is closer to the optical axis than an image-side end of the spatial frustum surface to the optical axis. The corresponding structure is disposed on the image-side surface of the first peripheral portion and correspondingly disposed on the frustum surface and the spatial frustum surface. The spatial layer is formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure are disposed at intervals. The other one of the space adjusting structures includes a frustum surface and a corresponding structure. The frustum surface is disposed on the object-side surface of the third peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is closer to the optical axis than an image-side end of the frustum surface to the optical axis. The corresponding structure is disposed on the image-side surface of the second peripheral portion

and correspondingly disposed on the frustum surface. When the imaging lens assembly is in a first environment, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the space adjusting structures is Gγ; when the imaging lens assembly is in a second environment, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the space adjusting structures is Gγ′; an abbe number of the second lens element is Vd, and the following conditions are satisfied: 3 μm≤Gγ′<Gγ≤38 μm; and 8≤Vd≤29. The first environment and the second environment are satisfied at least one of a temperature-dependent relation and a humidity-dependent relation: a temperature of the first environment being Ta, a temperature of the second environment being Tb, and the temperature-dependent relation satisfied: 6K≤|Ta−Tb|≤148K; and a relative humidity of the first environment being RHa, a relative humidity of the second environment being RHb, and the humidity-dependent relation satisfied: 7%≤|RHa−RHb|≤89%.
(7) According to one aspect of the present disclosure, an electronic device includes the imaging lens assembly of any one of the aforementioned aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a partial cross-sectional view of an imaging lens assembly according to the 1st embodiment of the present disclosure.
(2) FIG. **1**B is a schematic view of the imaging lens assembly according to the 1st embodiment in FIG. **1**A.
(3) FIG. **1**C is a schematic view of the space adjusting structure under the first environment according to the 1st embodiment in FIG. **1**B.
(4) FIG. **1**D is a schematic view of the space adjusting structure under the second environment according to the 1st embodiment in FIG. **1**B.
(5) FIG. **1**E is a schematic view of the space adjusting structure under the first environment according to the 1st embodiment in FIG. **1**B.
(6) FIG. **1**F is a schematic view of the space adjusting structure under the second environment according to the 1st embodiment in FIG. **1**B.
(7) FIG. **1**G is an exploded view of the lens barrel and the lens element according to the 1st embodiment in FIG. **1**A.
(8) FIG. **1**H is an exploded view of the lens elements according to the 1st embodiment in FIG. **1**A.
(9) FIG. **1**I is a partial exploded view of the lens barrel according to the 1st embodiment in FIG. **1**A.
(10) FIG. **2**A is a partial cross-sectional view of an imaging lens assembly according to the 2nd embodiment of the present disclosure.
(11) FIG. **2**B is a schematic view of the imaging lens assembly according to the 2nd embodiment in FIG. **2**A.
(12) FIG. **2**C is a schematic view of the space adjusting structure under the first environment according to the 2nd embodiment in FIG. **2**B.
(13) FIG. **2**D is a schematic view of the space adjusting structure under the second environment according to the 2nd embodiment in FIG. **2**B.
(14) FIG. **2**E is an exploded view of the lens barrel and the lens element according to the 2nd embodiment in FIG. **2**A.
(15) FIG. **3**A is a partial cross-sectional view of an imaging lens assembly according to the 3rd embodiment of the present disclosure.
(16) FIG. **3**B is a schematic view of the imaging lens assembly according to the 3rd embodiment in FIG. **3**A.

(17) FIG. **3**C is a schematic view of the space adjusting structure under the first environment according to the 1st example of the 3rd embodiment in FIG. **3**B.
(18) FIG. **3**D is a schematic view of the space adjusting structure under the second environment according to the 1st example of the 3rd embodiment in FIG. **3**B.
(19) FIG. **3**E is a schematic view of the space adjusting structure under the first environment according to the 1st example of the 3rd embodiment in FIG. **3**B.
(20) FIG. **3**F is a schematic view of the space adjusting structure under the second environment according to the 1st example of the 3rd embodiment in FIG. **3**B.
(21) FIG. **3**G is an exploded view of the lens elements according to the 1st example of the 3rd embodiment in FIG. **3**A.
(22) FIG. **3**H is an exploded view of the lens elements according to the 1st example of the 3rd embodiment in FIG. **3**A.
(23) FIG. **3**I is a schematic view of the space adjusting structure under the first environment according to the 2nd example of the 3rd embodiment in FIG. **3**B.
(24) FIG. **3**J is a schematic view of the space adjusting structure under the second environment according to the 2nd example of the 3rd embodiment in FIG. **3**B.
(25) FIG. **3**K is a schematic view of the space adjusting structure under the first environment according to the 2nd example of the 3rd embodiment in FIG. **3**B.
(26) FIG. **3**L is a schematic view of the space adjusting structure under the second environment according to the 2nd example of the 3rd embodiment in FIG. **3**B.
(27) FIG. **4**A is a schematic view of a vehicle instrument according to the 4th embodiment of the present disclosure.
(28) FIG. **4**B is another schematic view of the vehicle instrument according to the 4th embodiment in FIG. **4**A.
(29) FIG. **4**C is still another schematic view of the vehicle instrument according to the 4th embodiment in FIG. **4**A.
(30) FIG. **4**D is another schematic view of the vehicle instrument according to the 4th embodiment in FIG. **4**A.
(31) FIG. **5**A is a schematic view of an electronic device according to the 5th embodiment of the present disclosure.
(32) FIG. **5**B is a block diagram of the electronic device according to the 5th embodiment in FIG. **5**A.

DETAILED DESCRIPTION

(33) The present disclosure provides an imaging lens assembly, and the imaging lens assembly includes a first lens element, a second lens element, a lens barrel and at least one space adjusting structure, wherein an optical axis passes through the imaging lens assembly. The first lens element includes a first optical effective portion and a first peripheral portion, wherein the optical axis passes through the first optical effective portion, and the first peripheral portion is disposed around the first optical effective portion. The second lens element is disposed on an image side of the first lens element, and includes a second optical effective portion and a second peripheral portion, wherein the optical axis passes through the second optical effective portion, the second peripheral portion is disposed around the second optical effective portion, and an object-side surface of the second peripheral portion is directly contacted with an image-side surface of the first peripheral portion. The lens barrel includes a cylindrical portion and a plate portion, wherein the cylindrical portion surrounds the optical axis with the optical axis as an axis, the plate portion is connected to the cylindrical portion and extends towards a direction close to the optical axis to form a light through hole, an accommodating space is formed via the cylindrical portion and the plate portion, the first lens element is disposed in the accommodating space, and an image-side surface of the plate portion is directly contacted with an object-side surface of the first peripheral portion.
(34) Further, the first environment and the second environment are satisfied at least one of a

temperature-dependent relation and a humidity-dependent relation: a temperature of the first environment is Ta, a temperature of the second environment is Tb, and the temperature-dependent relation is satisfied: $6K \leq |Ta-Tb| \leq 148K$; and a relative humidity of the first environment is RHa, a relative humidity of the second environment is RHb, and the humidity-dependent relation is satisfied: $7\% \leq |RHa-RHb| \leq 89\%$.

(35) The space adjusting structure can be formed via the first peripheral portion of the first lens element and the plate portion of the lens barrel, and the space adjusting structure includes a frustum surface, a spatial frustum surface, a corresponding structure and a spatial layer. The frustum surface is disposed on the object-side surface of the first peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is closer to the optical axis than an image-side end of the frustum surface to the optical axis. The spatial frustum surface is disposed on the object-side surface of the first peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface is farther from the optical axis than an image-side end of the spatial frustum surface from the optical axis. The corresponding structure is disposed on the image-side surface of the plate portion and correspondingly disposed on the frustum surface and the spatial frustum surface. The spatial layer is formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure are disposed at intervals. When the imaging lens assembly is in a first environment, the frustum surface and the corresponding structure can be directly contacted, and a minimum spacing distance between the spatial frustum surface and the corresponding structure is G; when the imaging lens assembly is in a second environment, the frustum surface and the corresponding structure can be disposed at intervals, and the minimum spacing distance between the spatial frustum surface and the corresponding structure is G′; on a cross section along the optical axis, and an angle between the frustum surface and the spatial frustum surface is δ, the following conditions can be satisfied: $0\ \mu m \leq G' < G \leq 37\ \mu m$; and 18 degrees $\leq \theta \leq$ 130 degrees.

(36) The expansion rate between the optical elements is different due to the environmental variety, and the cushion space between the optical elements can be provided by changing the dimension of the spatial layers. Moreover, the interference between the optical elements after the variety of the environmental condition can be reduced via the structural design of the aforementioned space adjusting structures, so as to avoid the stress generated owing to the interference to lead the deformation of the optical lens elements. In other words, the deformation of the lens elements generated owing to the stress can be avoided, so as to maintain the stability of the optical quality, wherein the variety of the environmental condition can be the variety of temperature or the variety of humidity. When the expansion rate of the first lens element about the environmental variety is smaller than the expansion rate of the lens barrel about the environmental variety, the interference owing to the expansion can be avoided via the aforementioned structure.

(37) Moreover, the assembling positioning between the lens barrel and the first lens element can be enhanced under the first environment by the direct contact between the frustum surface and the corresponding structure; the interference between the frustum surface and the corresponding structure can be avoided under the second environment by the frustum surface and the corresponding structure disposed at intervals to form the spatial layer.

(38) Further, a number of the space adjusting structure can be two, wherein one of the space adjusting structures is formed via the first peripheral portion of the first lens element and the plate portion of the lens barrel, and the other one of the space adjusting structures is formed via the first peripheral portion of the first lens element and the second peripheral portion of the second lens element.

(39) The other one of the space adjusting structures includes a frustum surface, a spatial frustum surface, a corresponding structure and a spatial layer. The frustum surface is disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is farther from the optical axis than an image-side end of the

frustum surface from the optical axis. The spatial frustum surface is disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface is closer to the optical axis than an image-side end of the spatial frustum surface to the optical axis. The corresponding structure is disposed on the image-side surface of the first peripheral portion and correspondingly disposed on the frustum surface and the spatial frustum surface. The spatial layer is formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure are disposed at intervals.
(40) When the imaging lens assembly is in a first environment, the frustum surface and the corresponding structure of the one of the space adjusting structures can be directly contacted, the frustum surface and the corresponding structure of the other one of the space adjusting structures can be directly contacted, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the space adjusting structures is $G\alpha$, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the space adjusting structures is $G\beta$; when the imaging lens assembly is in a second environment, the frustum surface and the corresponding structure of the one of the space adjusting structures can be disposed at intervals, the frustum surface and the corresponding structure of the other one of the space adjusting structures can be disposed at intervals, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the space adjusting structures is $G\alpha'$, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the space adjusting structures is $G\beta'$; on a cross section along the optical axis, an angle between the frustum surface and the spatial frustum surface of the one of the space adjusting structures is $\theta\alpha$, and an angle between the frustum surface and the spatial frustum surface of the other one of the space adjusting structures is $\theta\beta$, the following conditions can be satisfied: $0\ \mu m \leq G\alpha' < G\alpha \leq 37\ \mu m$; $0\ \mu m \leq G\beta' < G\beta \leq 38\ \mu m$; 18 degrees$\leq\theta\alpha\leq$130 degrees; and 18 degrees$\leq\theta\beta\leq$130 degrees.
(41) When the expansion rate of the first lens element about the environmental variety is smaller than the expansion rate of the lens barrel about the environmental variety and the expansion rate of the second lens element about the environmental variety, so as to obtain the better effect of avoiding interference via the aforementioned structure, the interference owing to the expansion can be avoided, and the assembling stability can be maintained.
(42) When $\theta\alpha$ satisfies the aforementioned condition, the stressed direction which the first lens element and the corresponding structure of the one of the space adjusting structures are directly contacted can be dispersed, so as to avoid the deformation of the first lens element owing to stressed.
(43) When $\theta\beta$ satisfies the aforementioned condition, the stressed direction which the second lens element and the corresponding structure of the other one of the space adjusting structures are directly contacted can be dispersed, so as to avoid the deformation of the second lens element owing to stressed.
(44) Furthermore, the assembling positioning between the lens barrel and the first lens element can be enhanced under the first environment by the direct contact between the frustum surface and the corresponding structure of the one of the space adjusting structures; by the disposition at intervals of the frustum surface and the corresponding structure of the one of the space adjusting structures, the spatial layer can be formed between the frustum surface and the corresponding structure of the one of the space adjusting structures under the second environment to avoid the interference; the assembling positioning between the first lens element and the second lens element can be enhanced under the first environment by the direct contact between the frustum surface and the corresponding structure of the other one of the space adjusting structures; by the disposition at intervals of the frustum surface and the corresponding structure of the other one of the space adjusting structures, the spatial layer can be formed between the frustum surface and the

corresponding structure of the other one of the space adjusting structures under the second environment to avoid the interference.

(45) The imaging lens assembly can further include a third lens element, wherein the third lens element is disposed on an image side of the second lens element. The third lens element includes a third optical effective portion and a third peripheral portion, wherein the optical axis passes through the third optical effective portion, the third peripheral portion is disposed around the third optical effective portion, and an object-side surface of the third peripheral portion is directly contacted with an image-side surface of the second peripheral portion.

(46) Or, a number of the space adjusting structure can be two, wherein the one of the space adjusting structures is formed via the first peripheral portion of the first lens element and the second peripheral portion of the second lens element, and the other one of the space adjusting structures is formed via the second peripheral portion of the second lens element and the third peripheral portion of the third lens element.

(47) The one of the space adjusting structures includes a frustum surface, a spatial frustum surface, a corresponding structure and a spatial layer. The frustum surface is disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is farther from the optical axis than an image-side end of the frustum surface from the optical axis. The spatial frustum surface is disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface is closer to the optical axis than an image-side end of the spatial frustum surface to the optical axis. The corresponding structure is disposed on the image-side surface of the first peripheral portion and correspondingly disposed on the frustum surface and the spatial frustum surface. The spatial layer is formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure are disposed at intervals.

(48) The other one of the space adjusting structures includes a frustum surface and a corresponding structure. The frustum surface is disposed on the object-side surface of the third peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface is closer to the optical axis than an image-side end of the frustum surface to the optical axis. The corresponding structure is disposed on the image-side surface of the second peripheral portion and correspondingly disposed on the frustum surface.

(49) When the imaging lens assembly is in the first environment, the frustum surface and the corresponding structure of the one of the space adjusting structures can be directly contacted, and a minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the space adjusting structures is $G\gamma$; when the imaging lens assembly is in the second environment, the frustum surface and the corresponding structure of the one of the space adjusting structures can be disposed at intervals, and the minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the space adjusting structures is $G\gamma'$; on the cross section along the optical axis, an angle between the frustum surface and the spatial frustum surface of the one of the space adjusting structures is $\theta\gamma$, the following conditions can be satisfied: $3\ \mu m \leq G\gamma' < G\gamma \leq 38\ \mu m$; and $18\ \text{degrees} \leq \theta\gamma \leq 130\ \text{degrees}$.

(50) When the expansion rate of the second lens element about the environmental variety is larger than the expansion rate of the first lens element about the environmental variety and the expansion rate of the third lens element about the environmental variety, the interference owing to the expansion can be avoided via the aforementioned structure.

(51) The other one of the space adjusting structures can further include a spatial frustum surface and a spatial layer. The spatial frustum surface is disposed on the object-side surface of the third peripheral portion and disposed around the optical axis, an object-side end of the spatial frustum surface is farther from the optical axis than an image-side end of the spatial frustum surface from the optical axis, and the corresponding structure is further correspondingly disposed on the spatial

frustum surface. The spatial layer is formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure are disposed at intervals.

(52) When the imaging lens assembly is in a first environment, the frustum surface and the corresponding structure of the other one of the space adjusting structures can be directly contacted, and a minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the space adjusting structures is Gδ; when the imaging lens assembly is in a second environment, the frustum surface and the corresponding structure of the other one of the space adjusting structures can be disposed at intervals, and the minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the space adjusting structures is Gδ′; on the cross section along the optical axis, an angle between the frustum surface and the spatial frustum surface of the other one of the space adjusting structures is θδ, the following conditions can be satisfied: 3 μm≤Gδ′<Gδ≤39 μm; and 18 degrees≤θδ≤130 degrees.

(53) When an abbe number of the second lens element is Vd, the following condition can be satisfied: 8≤Vd≤29. When Vd satisfies the aforementioned condition, the volume of the second lens element is easily changed owing to the environmental variety. Further, the following condition can be satisfied: 8≤Vd≤22. Further, the following condition can be satisfied: 8≤Vd≤20.5.

(54) The first peripheral portion can include a bearing surface vertical to the optical axis, and the bearing surface and the plate portion are directly contacted. Therefore, the axial assembling stability of the first lens element can be promoted. Moreover, the directly contact of the bearing surface can be maintained under the first environment and the second environment.

(55) The second peripheral portion can include a bearing surface vertical to the optical axis, and the bearing surface and the first peripheral portion are directly contacted. Therefore, the axial assembling stability of the second lens element can be promoted.

(56) A diameter of the first lens element can be smaller than a diameter of the second lens element, and the diameter of the second lens element can be smaller than a diameter of the third lens element. Therefore, the aforementioned structure can be favorable for the optical design of the imaging lens assembly.

(57) Each of the aforementioned features of the imaging lens assembly can be utilized in various combinations for achieving the corresponding effects.

(58) The present disclosure provides an electronic device, which includes the aforementioned imaging lens assembly.

(59) According to the aforementioned embodiment, specific embodiments and examples are provided, and illustrated via figures.

1st Embodiment

(60) FIG. **1**A is a partial cross-sectional view of an imaging lens assembly **100** according to the 1st embodiment of the present disclosure. FIG. **1**B is a schematic view of the imaging lens assembly **100** according to the 1st embodiment in FIG. **1**A. In FIGS. **1**A and **1**B, the imaging lens assembly **100** includes a plurality of lens elements **111**, **112**, **113**, **114**, **115**, **116**, an optical filter **117**, an image sensor **121**, a lens barrel **130** (labelled in FIG. **1**G) and two space adjusting structures **140**, **150**, wherein an optical axis X passes through the imaging lens assembly **100**, and the image sensor **121** is disposed on an image surface **122**.

(61) According to the 1st embodiment, the lens element **111** can be a first lens element, and the lens element **112** can be a second lens element. The first lens element includes a first optical effective portion **111***a* (labelled in FIG. **1**G) and a first peripheral portion **111***b*, wherein the optical axis X passes through the first optical effective portion **111***a*, and the first peripheral portion **111***b* is disposed around the first optical effective portion **111***a*. The second lens element is disposed on an image side of the first lens element, and includes a second optical effective portion **112***a* (labelled in FIG. **1**H) and a second peripheral portion **112***b*, wherein the optical axis X passes through the second optical effective portion **112***a*, the second peripheral portion **112***b* is disposed around the

second optical effective portion **112***a*, and an object-side surface of the second peripheral portion **112***b* is directly contacted with an image-side surface of the first peripheral portion **111***b*.

(62) The lens barrel **130** includes a cylindrical portion **131** and a plate portion **132**, wherein the cylindrical portion **131** surrounds the optical axis X with the optical axis X as an axis, the plate portion **132** is connected to the cylindrical portion **131** and extends towards a direction close to the optical axis X to form a light through hole **133**, and an accommodating space **134** (labelled in FIG. **1**G) is formed via the cylindrical portion **131** and the plate portion **132**. Moreover, the lens elements **111**, **112**, **113**, **114**, **115**, **116** are disposed in the accommodating space **134**, and an image-side surface of the plate portion **132** is directly contacted with an object-side surface of the first peripheral portion **111***b*, wherein the lens element **116** is fixed on the cylindrical portion **131** via a glue A.

(63) FIG. **1**C is a schematic view of the space adjusting structure **140** under the first environment according to the 1st embodiment in FIG. **1**B. FIG. **1**D is a schematic view of the space adjusting structure **140** under the second environment according to the 1st embodiment in FIG. **1**B. FIG. **1**E is a schematic view of the space adjusting structure **150** under the first environment according to the 1st embodiment in FIG. **1**B. FIG. **1**F is a schematic view of the space adjusting structure **150** under the second environment according to the 1st embodiment in FIG. **1**B. FIG. **1**G is an exploded view of the lens barrel **130** and the lens element **111** according to the 1st embodiment in FIG. **1**A. FIG. **1**H is an exploded view of the lens elements **111**, **112** according to the 1st embodiment in FIG. **1**A. In FIGS. **1**C to **1**H, the space adjusting structure **140** is formed via the first peripheral portion **111***b* of the first lens element (that is, the lens element **111**) and the plate portion **132** of the lens barrel **130**, the space adjusting structure **150** is formed via the first peripheral portion **111***b* of the first lens element and the second peripheral portion **112***b* of the second lens element (that is, the lens element **112**).

(64) In FIGS. **1**C, **1**D and **1**G, the space adjusting structure **140** includes a frustum surface **141**, a spatial frustum surface **142**, a corresponding structure **143** and a spatial layer **144**. The frustum surface **141** is disposed on the object-side surface of the first peripheral portion **111***b* and disposed around the optical axis X, and an object-side end of the frustum surface **141** is closer to the optical axis X than an image-side end of the frustum surface **141** to the optical axis X. The spatial frustum surface **142** is disposed on the object-side surface of the first peripheral portion **111***b* and disposed around the optical axis X, and an object-side end of the spatial frustum surface **142** is farther from the optical axis X than an image-side end of the spatial frustum surface **142** from the optical axis X. The corresponding structure **143** is disposed on the image-side surface of the plate portion **132** and correspondingly disposed on the frustum surface **141** and the spatial frustum surface **142**. The spatial layer **144** is formed between the spatial frustum surface **142** and the corresponding structure **143**, so that the spatial frustum surface **142** and the corresponding structure **143** are disposed at intervals.

(65) In FIGS. **1**E, **1**F and **1**H, the space adjusting structure **150** includes a frustum surface **151**, a spatial frustum surface **152**, a corresponding structure **153** and a spatial layer **154**. The frustum surface **151** is disposed on the object-side surface of the second peripheral portion **112***b* and disposed around the optical axis X, and an object-side end of the frustum surface **151** is farther from the optical axis X than an image-side end of the frustum surface **151** from the optical axis X. The spatial frustum surface **152** is disposed on the object-side surface of the second peripheral portion **112***b* and disposed around the optical axis X, and an object-side end of the spatial frustum surface **152** is closer to the optical axis X than an image-side end of the spatial frustum surface **152** to the optical axis X. The corresponding structure **153** is disposed on the image-side surface of the first peripheral portion **111***b* and correspondingly disposed on the frustum surface **151** and the spatial frustum surface **152**. The spatial layer **154** is formed between the spatial frustum surface **152** and the corresponding structure **153**, so that the spatial frustum surface **152** and the corresponding structure **153** are disposed at intervals.

(66) In particular, the expansion rate between the optical elements is different due to the environmental variety, and the cushion space between the optical elements can be provided by changing the dimension of the spatial layers **144**, **154**, wherein the variety of the environmental condition can be the variety of temperature or the variety of humidity. Further, the interference between the optical elements after the variety of the environmental condition can be reduced via the structural design of the space adjusting structures **140**, **150**, so as to avoid the stress generated owing to the interference to lead the deformation of the optical lens elements. Therefore, the deformation of the lens elements generated owing to the stress can be avoided, so as to maintain the stability of the optical quality. Moreover, the expansion rate of the first lens element (that is, the lens element **111**) about the environmental variety is smaller than the expansion rate of the lens barrel **130** about the environmental variety and the expansion rate of the second lens element (that is, the lens element **112**) about the environmental variety, so as to obtain the better effect of avoiding interference via the aforementioned structure, the interference owing to the expansion can be avoided, and the stability of the optical quality can be maintained.
(67) In FIG. **1**D, the first peripheral portion **111***b* includes a bearing surface **111***c* vertical to the optical axis X, and the bearing surface **111***c* and the plate portion **132** are directly contacted. Therefore, the axial assembling stability of the first lens element can be promoted.
(68) FIG. **1**I is a partial exploded view of the lens barrel **130** according to the 1st embodiment in FIG. **1**A. In FIGS. **1**A and **1**I, the lens barrel **130** further includes a coil **135**, an elastic element **136** and a plurality of magnetic elements **137**, wherein the coil **135** and the magnetic elements **137** are correspondingly disposed, and the elastic element **136** is disposed between the cylindrical portion **131** and the optical filter **117**.
(69) In FIGS. **1**C and **1**D, a temperature Ta of the first environment is 293.1 K, a relative humidity RHa of the first environment is 30%, a temperature Tb of the second environment is 303.1K, and a relative humidity RHb of the second environment is 50%, wherein when the imaging lens assembly **100** is in the first environment, the frustum surface **141** and the corresponding structure **143** are directly contacted; when the imaging lens assembly **100** is in the second environment, the frustum surface **141** and the corresponding structure **143** are disposed at intervals. In particular, the assembling positioning between the lens barrel **130** and the first lens element (that is, the lens element **111**) can be enhanced under the first environment; the spatial layer **144** can be also formed between the frustum surface **141** and the corresponding structure **143**, so as to avoid the interference under the second environment. Moreover, the directly contact of the bearing surface **111***c* can be maintained under the first environment and the second environment. In other words, when the corresponding structure **143** and the frustum surface **141** are disposed at intervals and the corresponding structure **143** and the spatial frustum surface **142** are disposed at intervals, the positioning of the first lens element is maintained by clamping the bearing surface **111***c*.
(70) In FIGS. **1**E and **1**F, a temperature Ta of the first environment is 293.1 K, a relative humidity RHa of the first environment is 30%, a temperature Tb of the second environment is 293.1K, and a relative humidity RHb of the second environment is 85%, wherein when the imaging lens assembly **100** is in the first environment, the frustum surface **151** and the corresponding structure **153** are directly contacted; when the imaging lens assembly **100** is in the second environment, the frustum surface **151** and the corresponding structure **153** are disposed at intervals. In particular, the assembling positioning between the first lens element (that is, the lens element **111**) and the second lens element (that is, the lens element **112**) can be enhanced under the first environment; the spatial layer **154** can be also formed between the frustum surface **151** and the corresponding structure **153**, so as to avoid the interference under the second environment.
(71) It should be mentioned that the line segments of chain line in FIGS. **1**C to **1**F are configured to indicate the portion of directly contact.
(72) In FIGS. **1**C to **1**F, when the imaging lens assembly **100** is in the first environment, a minimum spacing distance between the spatial frustum surface **142** and the corresponding structure

143 of the one of the space adjusting structures (that is, the space adjusting structure **140**) is Ga, a minimum spacing distance between the spatial frustum surface **152** and the corresponding structure **153** of the other one of the space adjusting structures (that is, the space adjusting structure **150**) is Gβ; when the imaging lens assembly **100** is in the second environment, the minimum spacing distance between the spatial frustum surface **142** and the corresponding structure **143** of the one of the space adjusting structures is Gα′, the minimum spacing distance between the spatial frustum surface **152** and the corresponding structure **153** of the other one of the space adjusting structures is Gβ′; on a cross section along the optical axis X, an angle between the frustum surface **141** and the spatial frustum surface **142** of the one of the space adjusting structures is θα, an angle between the frustum surface **151** and the spatial frustum surface **152** of the other one of the space adjusting structures is θβ, and an abbe number of the second lens element (that is, the lens element **112**) is Vd, the following conditions of Table 1A are satisfied.

(73) TABLE-US-00001 TABLE 1A Gα (μm) 15 θα (degree) 40 Gα′ (μm) 5 θβ (degree) 40 Gβ (μm) 6 Vd 23.5 Gβ′ (μm) 0

2nd Embodiment

(74) FIG. **2**A is a partial cross-sectional view of an imaging lens assembly **200** according to the 2nd embodiment of the present disclosure. FIG. **2**B is a schematic view of the imaging lens assembly **200** according to the 2nd embodiment in FIG. **2**A. In FIGS. **2**A and **2**B, the imaging lens assembly **200** includes a plurality of lens elements **211**, **212**, **213**, **214**, **215**, **216**, an optical filter **217**, an image sensor **221**, a lens barrel **230** (labelled in FIG. **2**E) and a space adjusting structure **240**, wherein an optical axis X passes through the imaging lens assembly **200**, and the image sensor **221** is disposed on an image surface **222**.

(75) According to the 2nd embodiment, the lens element **211** can be a first lens element. The first lens element includes a first optical effective portion **211***a* (labelled in FIG. **2**E) and a first peripheral portion **211***b*, wherein the optical axis X passes through the first optical effective portion **211***a*, and the first peripheral portion **211***b* is disposed around the first optical effective portion **211***a*.

(76) The lens barrel **230** includes a cylindrical portion **231** and a plate portion **232**, wherein the cylindrical portion **231** surrounds the optical axis X with the optical axis X as an axis, the plate portion **232** is connected to the cylindrical portion **231** and extends towards a direction close to the optical axis X to form a light through hole **233**, and an accommodating space **234** (labelled in FIG. **2**E) is formed via the cylindrical portion **231** and the plate portion **232**. Moreover, the lens elements **211**, **212**, **213**, **214**, **215**, **216** are disposed in the accommodating space **234**, and an image-side surface of the plate portion **232** is directly contacted with an object-side surface of the first peripheral portion **211***b*.

(77) FIG. **2**C is a schematic view of the space adjusting structure **240** under the first environment according to the 2nd embodiment in FIG. **2**B. FIG. **2**D is a schematic view of the space adjusting structure **240** under the second environment according to the 2nd embodiment in FIG. **2**B. FIG. **2**E is an exploded view of the lens barrel **230** and the lens element **211** according to the 2nd embodiment in FIG. **2**A. In FIGS. **2**C to **2**E, the space adjusting structure **240** is formed via the first peripheral portion **211***b* of the first lens element (that is, the lens element **211**) and the plate portion **232** of the lens barrel **230**.

(78) Furthermore, the space adjusting structure **240** includes a frustum surface **241**, a spatial frustum surface **242**, a corresponding structure **243** and a spatial layer **244**. The frustum surface **241** is disposed on the object-side surface of the first peripheral portion **211***b* and disposed around the optical axis X, and an object-side end of the frustum surface **241** is closer to the optical axis X than an image-side end of the frustum surface **241** to the optical axis X. The spatial frustum surface **242** is disposed on the object-side surface of the first peripheral portion **211***b* and disposed around the optical axis X, and an object-side end of the spatial frustum surface **242** is farther from the optical axis X than an image-side end of the spatial frustum surface **242** from the optical axis X.

The corresponding structure **243** is disposed on the image-side surface of the plate portion **232** and correspondingly disposed on the frustum surface **241** and the spatial frustum surface **242**. The spatial layer **244** is formed between the spatial frustum surface **242** and the corresponding structure **243**, so that the spatial frustum surface **242** and the corresponding structure **243** are disposed at intervals.

(79) In particular, the interference between the optical elements after the variety of the environmental condition can be reduced via the structural design of the space adjusting structure **240**. Therefore, the stress generated owing to the interference to lead the deformation of the optical lens elements can be avoided, and the stability of the optical quality can be further maintained, wherein the variety of the environmental condition can be the variety of temperature or the variety of humidity. Further, when the expansion rate of the first lens element (that is, the lens element **211**) about the environmental variety is smaller than the expansion rate of the lens barrel **230** about the environmental variety, the interference owing to the expansion can be avoided via the aforementioned structure.

(80) In FIGS. **2**C and **2**D, a temperature Ta of the first environment is 293.1 K, a relative humidity RHa of the first environment is 30%, a temperature Tb of the second environment is 358.1K, and a relative humidity RHb of the second environment is 85%, wherein when the imaging lens assembly **200** is in the first environment, the frustum surface **241** and the corresponding structure **243** are directly contacted; when the imaging lens assembly **200** is in the second environment, the frustum surface **241** and the corresponding structure **243** are disposed at intervals. In particular, the assembling positioning between the lens barrel **230** and the first lens element (that is, the lens element **211**) can be enhanced under the first environment; the spatial layer **244** can be also formed between the frustum surface **241** and the corresponding structure **243**, so as to avoid the interference under the second environment.

(81) It should be mentioned that the line segments of chain line in FIGS. **2**C and **2**D are configured to indicate the portion of directly contact.

(82) In FIGS. **2**C and **2**D, when the imaging lens assembly **200** is in the first environment, a minimum spacing distance between the spatial frustum surface **242** and the corresponding structure **243** is G; when the imaging lens assembly **200** is in the second environment, the minimum spacing distance between the spatial frustum surface **242** and the corresponding structure **243** is G′; on a cross section along the optical axis X, an angle between the frustum surface **241** and the spatial frustum surface **242** is θ, the following conditions of Table 2A are satisfied.

(83) TABLE-US-00002 TABLE 2A G (μm) 15 θ (degree) 50 G′ (μm) 0

(84) Further, all of other structures and dispositions according to the 2nd embodiment are the same as the structures and the dispositions according to the 1st embodiment, and will not be described again herein.

3rd Embodiment

(85) FIG. **3**A is a partial cross-sectional view of an imaging lens assembly **300** according to the 3rd embodiment of the present disclosure. FIG. **3**B is a schematic view of the imaging lens assembly **300** according to the 3rd embodiment in FIG. **3**A. In FIGS. **3**A and **3**B, the imaging lens assembly **300** includes a plurality of lens elements **311**, **312**, **313**, **314**, **315**, **316**, **317**, an optical filter **318**, an image sensor **321**, a lens barrel **330** and a plurality of space adjusting structures **340**, **350**, **360**, **370**, wherein an optical axis X passes through the imaging lens assembly **300**, and the image sensor **321** is disposed on an image surface **322**.

(86) FIG. **3**C is a schematic view of the space adjusting structure **340** under the first environment according to the 1st example of the 3rd embodiment in FIG. **3**B. FIG. **3**D is a schematic view of the space adjusting structure **340** under the second environment according to the 1st example of the 3rd embodiment in FIG. **3**B. FIG. **3**E is a schematic view of the space adjusting structure **350** under the first environment according to the 1st example of the 3rd embodiment in FIG. **3**B. FIG. **3**F is a schematic view of the space adjusting structure **350** under the second environment according to the

1st example of the 3rd embodiment in FIG. **3**B. FIG. **3**G is an exploded view of the lens elements **311**, **312** according to the 1st example of the 3rd embodiment in FIG. **3**A. FIG. **3**H is an exploded view of the lens elements **312**, **313** according to the 1st example of the 3rd embodiment in FIG. **3**A. In FIGS. **3**B to **3**H, according to the 1st example of the 3rd embodiment, the lens element **311** can be a first lens element, the lens element **312** can be a second lens element, and the lens element **313** can be a third lens element, wherein a diameter of the first lens element is smaller than a diameter of the second lens element, and the diameter of the second lens element is smaller than a diameter of the third lens element.

(87) The first lens element includes a first optical effective portion **311***a* and a first peripheral portion **311***b*, wherein the optical axis X passes through the first optical effective portion **311***a*, and the first peripheral portion **311***b* is disposed around the first optical effective portion **311***a*. The second lens element is disposed on an image side of the first lens element, and includes a second optical effective portion **312***a* and a second peripheral portion **312***b*, wherein the optical axis X passes through the second optical effective portion **312***a*, the second peripheral portion **312***b* is disposed around the second optical effective portion **312***a*, and an object-side surface of the second peripheral portion **312***b* is directly contacted with an image-side surface of the first peripheral portion **311***b*. The third lens element is disposed on an image side of the second lens element, and includes a third optical effective portion **313***a* and a third peripheral portion **313***b*, wherein the optical axis X passes through the third optical effective portion **313***a*, the third peripheral portion **313***b* is disposed around the third optical effective portion **313***a*, and an object-side surface of the third peripheral portion **313***b* is directly contacted with an image-side surface of the second peripheral portion **312***b*.

(88) In FIGS. **3**C, **3**D and **3**G, the space adjusting structure **340** is formed via the first peripheral portion **311***b* of the first lens element and the second peripheral portion **312***b* of the second lens element, and the space adjusting structure **340** includes a frustum surface **341**, a spatial frustum surface **342**, a corresponding structure **343** and a spatial layer **344**. The frustum surface **341** is disposed on the object-side surface of the second peripheral portion **312***b* and disposed around the optical axis X, and an object-side end of the frustum surface **341** is farther from the optical axis X than an image-side end of the frustum surface **341** from the optical axis X. The spatial frustum surface **342** is disposed on the object-side surface of the second peripheral portion **312***b* and disposed around the optical axis X, and an object-side end of the spatial frustum surface **342** is closer to the optical axis X than an image-side end of the spatial frustum surface **342** to the optical axis X. The corresponding structure **343** is disposed on the image-side surface of the first peripheral portion **311***b* and correspondingly disposed on the frustum surface **341** and the spatial frustum surface **342**. The spatial layer **344** is formed between the spatial frustum surface **342** and the corresponding structure **343**, so that the spatial frustum surface **342** and the corresponding structure **343** are disposed at intervals.

(89) In FIGS. **3**E, **3**F and **3**H, the space adjusting structure **350** is formed via the second peripheral portion **312***b* of the second lens element and the third peripheral portion **313***b* of the third lens element, wherein the space adjusting structure **350** includes a frustum surface **351**, a spatial frustum surface **352** and a corresponding structure **353**. The frustum surface **351** is disposed on the object-side surface of the third peripheral portion **313***b* and disposed around the optical axis X, and an object-side end of the frustum surface **351** is closer to the optical axis X than an image-side end of the frustum surface **351** to the optical axis X. The spatial frustum surface **352** is disposed on the object-side surface of the third peripheral portion **313***b* and disposed around the optical axis X, and an object-side end of the spatial frustum surface **352** is farther from the optical axis X than an image-side end of the spatial frustum surface **352** from the optical axis X. The corresponding structure **353** is disposed on the image-side surface of the second peripheral portion **312***b* and correspondingly disposed on the frustum surface **351**.

(90) In particular, the interference between the optical elements after the variety of the

environmental condition can be reduced via the structural design of the space adjusting structures **340**, **350**. Therefore, the stress generated owing to the interference to lead the deformation of the optical lens elements can be avoided, and the stability of the optical quality can be further maintained, wherein the variety of the environmental condition can be the variety of temperature or the variety of humidity. Further, when the expansion rate of the second lens element about the environmental variety is larger than the expansion rate of the first lens element about the environmental variety and the expansion rate of the third lens element about the environmental variety, the interference owing to the expansion can be avoided via the aforementioned structure.
(91) In FIGS. **3**C and **3**D, the second peripheral portion **312***b* includes a bearing surface **312***c* vertical to the optical axis X, and the bearing surface **312***c* and the first peripheral portion **311***b* are directly contacted. Therefore, the axial assembling stability of the second lens element can be promoted.
(92) In FIGS. **3**C and **3**D, a temperature Ta of the first environment is 293.1 K, a relative humidity RHa of the first environment is 50%, a temperature Tb of the second environment is 358.1K, and a relative humidity RHb of the second environment is 50%, wherein when the imaging lens assembly **300** is in the first environment, the frustum surface **341** and the corresponding structure **343** are directly contacted; when the imaging lens assembly **300** is in the second environment, the frustum surface **341** and the corresponding structure **343** are disposed at intervals.
(93) In FIGS. **3**E and **3**F, a temperature Ta of the first environment is 263.1 K, a relative humidity RHa of the first environment is 50%, a temperature Tb of the second environment is 358.1K, and a relative humidity RHb of the second environment is 35%, wherein when the imaging lens assembly **300** is in the first environment, the frustum surface **351** and the corresponding structure **353** are directly contacted; when the imaging lens assembly **300** is in the second environment, the frustum surface **351** and the corresponding structure **353** are disposed at intervals.
(94) It should be mentioned that the line segments of chain line in FIGS. **3**C to **3**F are configured to indicate the portion of directly contact.
(95) In FIGS. **3**C to **3**F, when the imaging lens assembly **300** is in the first environment, a minimum spacing distance between the spatial frustum surface **342** and the corresponding structure **343** of the one of the space adjusting structures (that is, the space adjusting structure **340**) is Gγ; when the imaging lens assembly **300** is in a second environment, the minimum spacing distance between the spatial frustum surface **342** and the corresponding structure **343** of the one of the space adjusting structures is Gγ′; on a cross section along the optical axis X, an angle between the frustum surface **341** and the spatial frustum surface **342** of the one of the space adjusting structures is θγ, an angle between the frustum surface **351** and the spatial frustum surface **352** of the other one of the space adjusting structures (that is, the space adjusting structure **350**) is θδ, and an abbe number of the second lens element (that is, the lens element **312**) is Vd, the following conditions of Table 3A are satisfied.
(96) TABLE-US-00003 TABLE 3A Gγ (μm) 12 θγ (degree) 50 Gγ′ (μm) 4 θδ (degree) 55 Vd 18.4
(97) FIG. **3**I is a schematic view of the space adjusting structure **360** under the first environment according to the 2nd example of the 3rd embodiment in FIG. **3**B. FIG. **3**J is a schematic view of the space adjusting structure **360** under the second environment according to the 2nd example of the 3rd embodiment in FIG. **3**B. FIG. **3**K is a schematic view of the space adjusting structure **370** under the first environment according to the 2nd example of the 3rd embodiment in FIG. **3**B. FIG. **3**L is a schematic view of the space adjusting structure **370** under the second environment according to the 2nd example of the 3rd embodiment in FIG. **3**B. In FIGS. **3**I to **3**L, according to the 2nd example of the 3rd embodiment, the lens element **313** can be a first lens element, the lens element **314** can be a second lens element, and the lens element **315** can be a third lens element, wherein a diameter of the first lens element is smaller than a diameter of the second lens element, and the diameter of the second lens element is smaller than a diameter of the third lens element.
(98) The first lens element includes a first optical effective portion (its reference numeral is

omitted) and a first peripheral portion **313***d*, wherein the optical axis X passes through the first optical effective portion, and the first peripheral portion **313***d* is disposed around the first optical effective portion. The second lens element is disposed on an image side of the first lens element, and includes a second optical effective portion (its reference numeral is omitted) and a second peripheral portion **314***b*, wherein the optical axis X passes through the second optical effective portion, the second peripheral portion **314***b* is disposed around the second optical effective portion, and an object-side surface of the second peripheral portion **314***b* is directly contacted with an image-side surface of the first peripheral portion **313***d*. The third lens element is disposed on an image side of the second lens element, and includes a third optical effective portion (its reference numeral is omitted) and a third peripheral portion **315***b*, wherein the optical axis X passes through the third optical effective portion, the third peripheral portion **315***b* is disposed around the third optical effective portion, and an object-side surface of the third peripheral portion **315***b* is directly contacted with an image-side surface of the second peripheral portion **314***b*.
(99) In FIGS. **31** and **3**J, the space adjusting structure **360** is formed via the first peripheral portion **313***d* of the first lens element and the second peripheral portion **314***b* of the second lens element, and the space adjusting structure **360** includes a frustum surface **361**, a spatial frustum surface **362**, a corresponding structure **363** and a spatial layer **364**. The frustum surface **361** is disposed on the object-side surface of the second peripheral portion **314***b* and disposed around the optical axis X, and an object-side end of the frustum surface **361** is farther from the optical axis X than an image-side end of the frustum surface **361** from the optical axis X. The spatial frustum surface **362** is disposed on the object-side surface of the second peripheral portion **314***b* and disposed around the optical axis X, and an object-side end of the spatial frustum surface **362** is closer to the optical axis X than an image-side end of the spatial frustum surface **362** to the optical axis X. The corresponding structure **363** is disposed on the image-side surface of the first peripheral portion **313***d* and correspondingly disposed on the frustum surface **361** and the spatial frustum surface **362**. The spatial layer **364** is formed between the spatial frustum surface **362** and the corresponding structure **363**, so that the spatial frustum surface **362** and the corresponding structure **363** are disposed at intervals.
(100) In FIGS. **3**K and **3**L, the space adjusting structure **370** is formed via the second peripheral portion **314***b* of the second lens element and the third peripheral portion **315***b* of the third lens element, wherein the space adjusting structure **370** includes a frustum surface **371**, a spatial frustum surface **372**, a corresponding structure **373** and a spatial layer **374**. The frustum surface **371** is disposed on the object-side surface of the third peripheral portion **315***b* and disposed around the optical axis X, and an object-side end of the frustum surface **371** is closer to the optical axis X than an image-side end of the frustum surface **371** to the optical axis X. The spatial frustum surface **372** is disposed on the object-side surface of the third peripheral portion **315***b* and disposed around the optical axis X, and an object-side end of the spatial frustum surface **372** is farther from the optical axis X than an image-side end of the spatial frustum surface **372** from the optical axis X. The corresponding structure **373** is disposed on the image-side surface of the second peripheral portion **314***b* and correspondingly disposed on the frustum surface **371** and the spatial frustum surface **372**. The spatial layer **374** is formed between the spatial frustum surface **372** and the corresponding structure **373**, so that the spatial frustum surface **372** and the corresponding structure **373** are disposed at intervals.
(101) In particular, the interference between the optical elements after the variety of the environmental condition can be reduced via the structural design of the space adjusting structures **360**, **370**. Therefore, the stress generated owing to the interference to lead the deformation of the optical lens elements can be avoided, and the stability of the optical quality can be further maintained. Further, when the expansion rate of the second lens element about the environmental variety is larger than the expansion rate of the first lens element about the environmental variety and the expansion rate of the third lens element about the environmental variety, the interference

owing to the expansion can be avoided via the aforementioned structure.
(102) In FIGS. **31** and **3**J, the second peripheral portion **314***b* includes a bearing surface **314***c* vertical to the optical axis X, and the bearing surface **314***c* and the first peripheral portion **313***d* are directly contacted. Therefore, the axial assembling stability of the second lens element can be promoted.
(103) In FIGS. **3**I and **3**J, a temperature Ta of the first environment is 293.1 K, a relative humidity RHa of the first environment is 30%, a temperature Tb of the second environment is 283.1K, and a relative humidity RHb of the second environment is 95%, wherein when the imaging lens assembly **300** is in the first environment, the frustum surface **361** and the corresponding structure **363** are directly contacted; when the imaging lens assembly **300** is in the second environment, the frustum surface **361** and the corresponding structure **363** are disposed at intervals.
(104) In FIGS. **3**K and **3**L, a temperature Ta of the first environment is 293.1 K, a relative humidity RHa of the first environment is 30%, a temperature Tb of the second environment is 358.1K, and a relative humidity RHb of the second environment is 50%, wherein when the imaging lens assembly **300** is in the first environment, the frustum surface **371** and the corresponding structure **373** are disposed at intervals; when the imaging lens assembly **300** is in the second environment, the frustum surface **371** and the corresponding structure **373** are disposed at intervals.
(105) It should be mentioned that the line segments of chain line in FIGS. **3**I to **3**L are configured to indicate the portion of directly contact.
(106) In FIGS. **3**I to **3**L, when the imaging lens assembly **300** is in the first environment, a minimum spacing distance between the spatial frustum surface **362** and the corresponding structure **363** of the one of the space adjusting structures (that is, the space adjusting structure **360**) is Gγ, a minimum spacing distance between the spatial frustum surface **372** and the corresponding structure **373** of the other one of the space adjusting structures (that is, the space adjusting structure **370**) is Go; when the imaging lens assembly **300** is in a second environment, the minimum spacing distance between the spatial frustum surface **362** and the corresponding structure **363** of the one of the space adjusting structures is Gγ′, the minimum spacing distance between the spatial frustum surface **372** and the corresponding structure **373** of the other one of the space adjusting structures is Gδ′; on a cross section along the optical axis X, an angle between the frustum surface **361** and the spatial frustum surface **362** of the one of the space adjusting structures is θγ, and an angle between the frustum surface **371** and the spatial frustum surface **372** of the other one of the space adjusting structures is θδ, the following conditions of Table 3B are satisfied.
(107) TABLE-US-00004 TABLE 3B Gγ (μm) 21 Gδ′ (μm) 5 Gγ′ (μm) 10 θγ (degree) 50 Gδ (μm) 13 θδ (degree) 55
(108) Further, all of other structures and dispositions according to the 3rd embodiment are the same as the structures and the dispositions according to the 1st embodiment, and will not be described again herein.
4th Embodiment
(109) FIG. **4**A is a schematic view of a vehicle instrument **40** according to the 4th embodiment of the present disclosure. FIG. **4**B is another schematic view of the vehicle instrument **40** according to the 4th embodiment in FIG. **4**A. FIG. **4**C is still another schematic view of the vehicle instrument **40** according to the 4th embodiment in FIG. **4**A. FIG. **4**D is another schematic view of the vehicle instrument **40** according to the 4th embodiment in FIG. **4**A. In FIGS. **4**A to **4**D, the vehicle instrument **40** includes a plurality of imaging lens assemblies **41**. According to the 4th embodiment, a number of the imaging lens assemblies **41** is six, but the present disclosure is not limited thereto. In particular, the imaging lens assembly can be one of the imaging lens assemblies according to the aforementioned 1st embodiment to the 3rd embodiment, but the present disclosure is not limited thereto.
(110) In FIGS. **4**A and **4**B, the imaging lens assemblies **41** are automotive imaging lens assemblies, two of the imaging lens assemblies **41** are located under rearview mirrors on a left side and a right

side, respectively, and the aforementioned imaging lens assemblies **41** are configured to capture the image information of a visual angle α. In particular, the visual angle α can satisfy the following condition: 40 degrees<α<90 degrees. Therefore, the image information in the regions of two lanes on the left side and the right side can be captured.

(111) In FIG. **4**B, another two of the imaging lens assemblies **41** can be disposed in the inner space of the vehicle instrument **40**. In particular, the aforementioned two imaging lens assemblies **41** are disposed on a location close to the rearview mirror inside the vehicle instrument **40** and a location close to the rear car window, respectively. Moreover, the imaging lens assemblies **41** can be further disposed on the rearview mirrors on the left side and the right side except the mirror surface, respectively, but the present disclosure is not limited thereto.

(112) In FIG. **4**C, another two of the imaging lens assemblies **41** can be disposed on a front end of the vehicle instrument **40** and a rear end of the vehicle instrument **40**, respectively. By disposing the imaging lens assemblies **41** on the front end and the rear end of the vehicle instrument **40** and under the rearview mirror on the left side of the vehicle instrument **40** and the right side of the vehicle instrument **40**, it is favorable for the drivers obtaining external space informations in addition to the driving seat, such as external space informations I**1**, I**2**, I**3**, I**4**, but the present disclosure is not limited thereto. Therefore, more visual angles can be provided to reduce the blind spot, so that the driving safety can be improved.

(113) In FIG. **4**D, the imaging lens assembly **41** disposed on the rearview mirror in the vehicle instrument **40** can be configured to capture an inner space information **15**, so as to enhance the driving safety. Generally speaking, when the vehicle instrument of the prior art is parked and exposed to the sun, the effect of temperature drift of the imaging lens assembly is caused via the high temperature in the vehicle instrument, and the imaging lens assembly is even damaged, so as to influence the driving safety. The interference stress caused between the optical elements owing to the variety of the environmental condition can be avoided by disposing the space adjusting structures of the imaging lens assemblies **41** of the present disclosure, and hence the stability can be still maintained under the environment of the severe variety of the temperature and the imaging quality can be maintained.

5th Embodiment

(114) FIG. **5**A is a schematic view of an electronic device **50** according to the 5th embodiment of the present disclosure. FIG. **5**B is a block diagram of the electronic device **50** according to the 5th embodiment in FIG. **5**A. In FIGS. **5**A and **5**B, the electronic device **50** is a smart phone, and includes an imaging lens assembly.

(115) According to the 5th embodiment, the electronic device **50** includes four imaging lens assemblies, and the imaging lens assemblies are a long-focal telephoto imaging lens assembly **511**, an ultra-wide-angle imaging lens assembly **512**, an ultra-long-focal telephoto imaging lens assembly **513** and a wide-angle main imaging lens assembly **514**, wherein a visual angle of the long-focal telephoto imaging lens assembly **511** is between 30 degrees and 60 degrees, a visual angle of the ultra-wide-angle imaging lens assembly **512** is between 93 degrees and 175 degrees, a visual angle of the ultra-long-focal telephoto imaging lens assembly **513** is between 5 degrees and 30 degrees, and a visual angle of the wide-angle main imaging lens assembly **514** is between 65 degrees and 90 degrees, but the present disclosure is not limited thereto. Moreover, the function of optical zoom of the electronic device **50** can be obtained by switching the imaging lens assemblies with the different visual angles. It should be mentioned that a lens cover **52** is only configured to indicate the long-focal telephoto imaging lens assembly **511**, the ultra-wide-angle imaging lens assembly **512**, the ultra-long-focal telephoto imaging lens assembly **513** and the wide-angle main imaging lens assembly **514** disposed in the electronic device **50**, and the schematic view is not configured to mean that the lens cover **52** is removable. In particular, the long-focal telephoto imaging lens assembly **511**, the ultra-wide-angle imaging lens assembly **512**, the ultra-long-focal telephoto imaging lens assembly **513** and the wide-angle main imaging lens assembly **514** can be

one of the imaging lens assemblies according to the aforementioned 1st embodiment to the 3rd embodiment, but the present disclosure is not limited thereto.

(116) The electronic device **50** further includes an image sensor **53** and a user interface **54**, wherein the image sensor **53** is disposed on an image surface (not shown) of the long-focal telephoto imaging lens assembly **511**, the ultra-wide-angle imaging lens assembly **512**, the ultra-long-focal telephoto imaging lens assembly **513** and the wide-angle main imaging lens assembly **514**, and the user interface **54** can be a touch screen or a display screen, but the present disclosure is not limited thereto.

(117) Moreover, users enter a shooting mode via the user interface **54** of the electronic device **50**. At this moment, the imaging light is gathered on the image sensor **53** via the long-focal telephoto imaging lens assembly **511**, the ultra-wide-angle imaging lens assembly **512**, the ultra-long-focal telephoto imaging lens assembly **513** and the wide-angle main imaging lens assembly **514**, and an electronic signal about an image is output to an image signal processor (ISP) **55**.

(118) To meet a specification of the electronic device **50**, the electronic device **50** can further include an optical anti-shake mechanism **56**, which can be an optical image stabilization (OIS). Furthermore, the electronic device **50** can further include at least one auxiliary optical element (its reference numeral is omitted) and at least one sensing element **57**. According to the 5th embodiment, the auxiliary optical element is a flash module **58** and a focusing assisting module **59**. The flash module **58** can be for compensating a color temperature, and the focusing assisting module **59** can be an infrared distance measurement component, a laser focus module, etc. The sensing element **57** can have functions for sensing physical momentum and kinetic energy, such as an accelerator, a gyroscope, a Hall Effect Element, to sense shaking or jitters applied by hands of the user or external environments. Accordingly, an auto-focusing mechanism and the optical anti-shake mechanism **56** disposed on the imaging lens assembly (that is, the long-focal telephoto imaging lens assembly **511**, the ultra-wide-angle imaging lens assembly **512**, the ultra-long-focal telephoto imaging lens assembly **513**, the wide-angle main imaging lens assembly **514**) of the electronic device **50** can be enhanced to achieve the superior image quality. Furthermore, the electronic device **50** according to the present disclosure can have a capturing function with multiple modes, such as taking optimized selfies, high dynamic range (HDR) under a low light condition, 4K resolution recording, etc. Furthermore, the users can visually see a captured image of the camera through the touch screen and manually operate the view finding range on the touch screen to achieve the autofocus function of what you see is what you get.

(119) Furthermore, the electronic device **50** can further include, but not be limited to, a display, a control unit, a storage unit, a random access memory (RAM), a read-only memory (ROM), or the combination thereof.

(120) The foregoing description, for purpose of explanation, has been described with reference to specific examples. It is to be noted that Tables show different data of the different examples; however, the data of the different examples are obtained from experiments. The examples were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various examples with various modifications as are suited to the particular use contemplated. The examples depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

Claims

1. An imaging lens assembly, an optical axis passing through the imaging lens assembly, and the imaging lens assembly comprising: a first lens element, comprising: a first optical effective portion, wherein the optical axis passes through the first optical effective portion; and a first peripheral

portion disposed around the first optical effective portion; a second lens element, disposed on an image side of the first lens element, and comprising: a second optical effective portion, wherein the optical axis passes through the second optical effective portion; and a second peripheral portion disposed around the second optical effective portion, and an object-side surface of the second peripheral portion directly contacted with an image-side surface of the first peripheral portion; a lens barrel, comprising: a cylindrical portion surrounding the optical axis with the optical axis as an axis; and a plate portion connected to the cylindrical portion, extending towards a direction close to the optical axis to form a light through hole, an accommodating space formed via the cylindrical portion and the plate portion, the first lens element and the second lens element disposed in the accommodating space, and an image-side surface of the plate portion directly contacted with an object-side surface of the first peripheral portion; and two space adjusting structures, wherein one of the two space adjusting structures is formed via the first peripheral portion of the first lens element and the plate portion of the lens barrel, the other one of the two space adjusting structures is formed via the first peripheral portion of the first lens element and the second peripheral portion of the second lens element; wherein the one of the two space adjusting structures comprises: a frustum surface disposed on the object-side surface of the first peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface closer to the optical axis than an image-side end of the frustum surface to the optical axis; a spatial frustum surface disposed on the object-side surface of the first peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface farther from the optical axis than an image-side end of the spatial frustum surface from the optical axis; a corresponding structure disposed on the image-side surface of the plate portion and correspondingly disposed on the frustum surface and the spatial frustum surface; and a spatial layer formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure disposed at intervals; wherein the other one of the two space adjusting structures comprises: a frustum surface disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface farther from the optical axis than an image-side end of the frustum surface from the optical axis; a spatial frustum surface disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface closer to the optical axis than an image-side end of the spatial frustum surface to the optical axis; a corresponding structure disposed on the image-side surface of the first peripheral portion and correspondingly disposed on the frustum surface and the spatial frustum surface; and a spatial layer formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure disposed at intervals; wherein when the imaging lens assembly is in a first environment, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the two space adjusting structures is $G\alpha$, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the two space adjusting structures is $G\beta$; when the imaging lens assembly is in a second environment, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the two space adjusting structures is $G\alpha'$, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the two space adjusting structures is $G\beta'$, and the following conditions are satisfied:

$0\ \mu m \leq G\alpha' < G\alpha \leq 37\ \mu m$; and

$0\ \mu m \leq G\beta' < G\beta \leq 38\ \mu m$; wherein the first environment and the second environment are satisfied at least one of a temperature-dependent relation and a humidity-dependent relation: a temperature of the first environment being Ta, a temperature of the second environment being Tb, and the temperature-dependent relation satisfied: $6K \leq |Ta - Tb| \leq 148K$; and a relative humidity of the first environment being RHa, a relative humidity of the second environment being RHb, and the humidity-dependent relation satisfied: $7\% \leq |RHa - RHb| \leq 89\%$.

2. The imaging lens assembly of claim 1, wherein an abbe number of the second lens element is Vd, and the following condition is satisfied:
$8 \leq Vd \leq 29$.
3. The imaging lens assembly of claim 1, wherein when the imaging lens assembly is in the first environment, the frustum surface and the corresponding structure of the one of the two space adjusting structures are directly contacted.
4. The imaging lens assembly of claim 3, wherein when the imaging lens assembly is in the second environment, the frustum surface and the corresponding structure of the one of the two space adjusting structures are disposed at intervals.
5. The imaging lens assembly of claim 1, wherein when the imaging lens assembly is in the first environment, the frustum surface and the corresponding structure of the other one of the two space adjusting structures are directly contacted.
6. The imaging lens assembly of claim 5, wherein when the imaging lens assembly is in the second environment, the frustum surface and the corresponding structure of the other one of the two space adjusting structures are disposed at intervals.
7. The imaging lens assembly of claim 1, wherein the first peripheral portion comprises a bearing surface vertical to the optical axis, and the bearing surface and the plate portion are directly contacted.
8. The imaging lens assembly of claim 1, wherein on a cross section along the optical axis, an angle between the frustum surface and the spatial frustum surface of the one of the two space adjusting structures is $\theta\alpha$, and the following condition is satisfied:
18 degrees $\leq \theta\alpha \leq$ 130 degrees.
9. The imaging lens assembly of claim 1, wherein on a cross section along the optical axis, an angle between the frustum surface and the spatial frustum surface of the other one of the two space adjusting structures is $\theta\beta$, and the following condition is satisfied:
18 degrees $\leq \theta\beta \leq$ 130 degrees.
10. An imaging lens assembly, an optical axis passing through the imaging lens assembly, and the imaging lens assembly comprising: a first lens element, comprising: a first optical effective portion, wherein the optical axis passes through the first optical effective portion; and a first peripheral portion disposed around the first optical effective portion; a lens barrel, comprising: a cylindrical portion surrounding the optical axis with the optical axis as an axis; and a plate portion connected to the cylindrical portion, extending towards a direction close to the optical axis to form a light through hole, an accommodating space formed via the cylindrical portion and the plate portion, the first lens element disposed in the accommodating space, and an image-side surface of the plate portion directly contacted with an object-side surface of the first peripheral portion; and a space adjusting structure, wherein the space adjusting structure is formed via the first peripheral portion of the first lens element and the plate portion of the lens barrel, and the space adjusting structure comprises: a frustum surface disposed on the object-side surface of the first peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface closer to the optical axis than an image-side end of the frustum surface to the optical axis; a spatial frustum surface disposed on the object-side surface of the first peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface farther from the optical axis than an image-side end of the spatial frustum surface from the optical axis; a corresponding structure disposed on the image-side surface of the plate portion and correspondingly disposed on the frustum surface and the spatial frustum surface; and a spatial layer formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure disposed at intervals; wherein when the imaging lens assembly is in a first environment, a minimum spacing distance between the spatial frustum surface and the corresponding structure is G; when the imaging lens assembly is in a second environment, the minimum spacing distance between the spatial frustum surface and the corresponding structure is

G', and the following condition is satisfied:
$0\ \mu m \le G' < G \le 37\ \mu m$; wherein the first environment and the second environment are satisfied at least one of a temperature-dependent relation and a humidity-dependent relation: a temperature of the first environment being Ta, a temperature of the second environment being Tb, and the temperature-dependent relation satisfied: $6K \le |Ta-Tb| \le 148K$; and a relative humidity of the first environment being RHa, a relative humidity of the second environment being RHb, and the humidity-dependent relation satisfied: $7\% \le |RHa-RHb| \le 89\%$.
11. The imaging lens assembly of claim 10, wherein when the imaging lens assembly is in the first environment, the frustum surface and the corresponding structure are directly contacted.
12. The imaging lens assembly of claim 11, wherein when the imaging lens assembly is in the second environment, the frustum surface and the corresponding structure are disposed at intervals.
13. The imaging lens assembly of claim 10, wherein the first peripheral portion comprises a bearing surface vertical to the optical axis, and the bearing surface and the plate portion are directly contacted.
14. The imaging lens assembly of claim 10, wherein on a cross section along the optical axis, an angle between the frustum surface and the spatial frustum surface is θ, and the following condition is satisfied:
$18\ \text{degrees} \le \theta \le 130\ \text{degrees}$.
15. An imaging lens assembly, an optical axis passing through the imaging lens assembly, and the imaging lens assembly comprising: a first lens element, comprising: a first optical effective portion, wherein the optical axis passes through the first optical effective portion; and a first peripheral portion disposed around the first optical effective portion; a second lens element, disposed on an image side of the first lens element, and comprising: a second optical effective portion, wherein the optical axis passes through the second optical effective portion; and a second peripheral portion disposed around the second optical effective portion, and an object-side surface of the second peripheral portion directly contacted with an image-side surface of the first peripheral portion; a third lens element, disposed on an image side of the second lens element, and comprising: a third optical effective portion, wherein the optical axis passes through the third optical effective portion; and a third peripheral portion disposed around the third optical effective portion, and an object-side surface of the third peripheral portion directly contacted with an image-side surface of the second peripheral portion; and two space adjusting structures, wherein one of the two space adjusting structures is formed via the first peripheral portion of the first lens element and the second peripheral portion of the second lens element, the other one of the two space adjusting structures is formed via the second peripheral portion of the second lens element and the third peripheral portion of the third lens element; wherein the one of the two space adjusting structures comprises: a frustum surface disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface farther from the optical axis than an image-side end of the frustum surface from the optical axis; a spatial frustum surface disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface closer to the optical axis than an image-side end of the spatial frustum surface to the optical axis; a corresponding structure disposed on the image-side surface of the first peripheral portion and correspondingly disposed on the frustum surface and the spatial frustum surface; and a spatial layer formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure disposed at intervals; wherein the other one of the two space adjusting structures comprises: a frustum surface disposed on the object-side surface of the third peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface closer to the optical axis than an image-side end of the frustum surface to the optical axis; a spatial frustum surface disposed on the object-side surface of the third peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface farther from the optical axis than an

image-side end of the spatial frustum surface from the optical axis; a corresponding structure disposed on the image-side surface of the second peripheral portion and correspondingly disposed on the frustum surface and the spatial frustum surface; and a spatial layer formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure disposed at intervals; wherein when the imaging lens assembly is in a first environment, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the two space adjusting structures is Gγ, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the two space adjusting structures is Gδ; when the imaging lens assembly is in a second environment, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the two space adjusting structures is Gγ′, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the other one of the two space adjusting structures is Gδ′; an abbe number of the second lens element is Vd, and the following conditions are satisfied:

$3\ \mu m \leq G\gamma' < G\gamma \leq 38\ \mu m$;

$3\ \mu m \leq G\delta' < G\delta \leq 39\ \mu m$; and

$8 \leq Vd \leq 29$; wherein the first environment and the second environment are satisfied at least one of a temperature-dependent relation and a humidity-dependent relation: a temperature of the first environment being Ta, a temperature of the second environment being Tb, and the temperature-dependent relation satisfied: $6K \leq |Ta-Tb| \leq 148K$; and a relative humidity of the first environment being RHa, a relative humidity of the second environment being RHb, and the humidity-dependent relation satisfied: $7\% \leq |RHa-RHb| \leq 89\%$.

16. The imaging lens assembly of claim 15, wherein the abbe number of the second lens element is Vd, and the following condition is satisfied:

$8 \leq Vd \leq 22$.

17. The imaging lens assembly of claim 16, wherein the abbe number of the second lens element is Vd, and the following condition is satisfied:

$8 \leq Vd \leq 20.5$.

18. The imaging lens assembly of claim 15, wherein when the imaging lens assembly is in the first environment, the frustum surface and the corresponding structure of the one of the two space adjusting structures are directly contacted.

19. The imaging lens assembly of claim 18, wherein when the imaging lens assembly is in the second environment, the frustum surface and the corresponding structure of the one of the two space adjusting structures are disposed at intervals.

20. The imaging lens assembly of claim 15, wherein when the imaging lens assembly is in the first environment, the frustum surface and the corresponding structure of the other one of the two space adjusting structures are directly contacted.

21. The imaging lens assembly of claim 20, wherein when the imaging lens assembly is in the second environment, the frustum surface and the corresponding structure of the other one of the two space adjusting structures are disposed at intervals.

22. The imaging lens assembly of claim 15, wherein the second peripheral portion comprises a bearing surface vertical to the optical axis, and the bearing surface and the first peripheral portion are directly contacted.

23. The imaging lens assembly of claim 15, wherein on a cross section along the optical axis, an angle between the frustum surface and the spatial frustum surface of the one of the two space adjusting structures is θγ, and the following condition is satisfied:

18 degrees≤θγ≤130 degrees.

24. The imaging lens assembly of claim 15, wherein on a cross section along the optical axis, an angle between the frustum surface and the spatial frustum surface of the other one of the two space adjusting structures is θδ, and the following condition is satisfied:

18 degrees≤θδ≤130 degrees.

25. The imaging lens assembly of claim 15, wherein a diameter of the first lens element is smaller than a diameter of the second lens element, and the diameter of the second lens element is smaller than a diameter of the third lens element.

26. An imaging lens assembly, an optical axis passing through the imaging lens assembly, and the imaging lens assembly comprising: a first lens element, comprising: a first optical effective portion, wherein the optical axis passes through the first optical effective portion; and a first peripheral portion disposed around the first optical effective portion; a second lens element, disposed on an image side of the first lens element, and comprising: a second optical effective portion, wherein the optical axis passes through the second optical effective portion; and a second peripheral portion disposed around the second optical effective portion, and an object-side surface of the second peripheral portion directly contacted with an image-side surface of the first peripheral portion; a third lens element, disposed on an image side of the second lens element, and comprising: a third optical effective portion, wherein the optical axis passes through the third optical effective portion; and a third peripheral portion disposed around the third optical effective portion, and an object-side surface of the third peripheral portion directly contacted with an image-side surface of the second peripheral portion; and two space adjusting structures, wherein one of the two space adjusting structures is formed via the first peripheral portion of the first lens element and the second peripheral portion of the second lens element, the other one of the two space adjusting structures is formed via the second peripheral portion of the second lens element and the third peripheral portion of the third lens element; wherein the one of the two space adjusting structures comprises: a frustum surface disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface farther from the optical axis than an image-side end of the frustum surface from the optical axis; a spatial frustum surface disposed on the object-side surface of the second peripheral portion and disposed around the optical axis, and an object-side end of the spatial frustum surface closer to the optical axis than an image-side end of the spatial frustum surface to the optical axis; a corresponding structure disposed on the image-side surface of the first peripheral portion and correspondingly disposed on the frustum surface and the spatial frustum surface; and a spatial layer formed between the spatial frustum surface and the corresponding structure, so that the spatial frustum surface and the corresponding structure disposed at intervals; wherein the other one of the two space adjusting structures comprises: a frustum surface disposed on the object-side surface of the third peripheral portion and disposed around the optical axis, and an object-side end of the frustum surface closer to the optical axis than an image-side end of the frustum surface to the optical axis; and a corresponding structure disposed on the image-side surface of the second peripheral portion and correspondingly disposed on the frustum surface; wherein when the imaging lens assembly is in a first environment, a minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the two space adjusting structures is Gγ; when the imaging lens assembly is in a second environment, the minimum spacing distance between the spatial frustum surface and the corresponding structure of the one of the two space adjusting structures is Gγ′; an abbe number of the second lens element is Vd, and the following conditions are satisfied:

$3\ \mu m \leq G\gamma' < G\gamma \leq 38\ \mu m$; and

$8 \leq Vd \leq 29$; wherein the first environment and the second environment are satisfied at least one of a temperature-dependent relation and a humidity-dependent relation: a temperature of the first environment being Ta, a temperature of the second environment being Tb, and the temperature-dependent relation satisfied: 6K≤|Ta−Tb|≤148K; and a relative humidity of the first environment being RHa, a relative humidity of the second environment being RHb, and the humidity-dependent relation satisfied: 7%≤|RHa−RHb|≤89%.

27. The imaging lens assembly of claim 26, wherein the abbe number of the second lens element is Vd, and the following condition is satisfied:

$8 \leq Vd \leq 22$.

28. The imaging lens assembly of claim 26, wherein the abbe number of the second lens element is Vd, and the following condition is satisfied:

$8 \leq Vd \leq 20.5$.

29. The imaging lens assembly of claim 26, wherein when the imaging lens assembly is in the first environment, the frustum surface and the corresponding structure of the one of the two space adjusting structures are directly contacted.

30. The imaging lens assembly of claim 29, wherein when the imaging lens assembly is in the second environment, the frustum surface and the corresponding structure of the one of the two space adjusting structures are disposed at intervals.

31. The imaging lens assembly of claim 26, wherein when the imaging lens assembly is in the first environment, the frustum surface and the corresponding structure of the other one of the two space adjusting structures are directly contacted.

32. The imaging lens assembly of claim 31, wherein when the imaging lens assembly is in the second environment, the frustum surface and the corresponding structure of the other one of the two space adjusting structures are disposed at intervals.

33. The imaging lens assembly of claim 26, wherein the second peripheral portion comprises a bearing surface vertical to the optical axis, and the bearing surface and the first peripheral portion are directly contacted.

34. The imaging lens assembly of claim 26, wherein on a cross section along the optical axis, an angle between the frustum surface and the spatial frustum surface of the one of the two space adjusting structures is $\theta\gamma$, and the following condition is satisfied:

18 degrees $\leq \theta\gamma \leq$ 130 degrees.

35. The imaging lens assembly of claim 26, wherein a diameter of the first lens element is smaller than a diameter of the second lens element, and the diameter of the second lens element is smaller than a diameter of the third lens element.

36. An electronic device, comprising: the imaging lens assembly of claim 1, 10, 15 or 26.